

Compulsory assignment, BAN404

From March 10, 2025 until March 17, 2025, 23.59

Introduction

Note that only one student in each group should deliver the assignment. Therefore, make sure to write your names and student numbers on it. The assignment should be submitted as a *zip-file* containing two files. The two files can either be

1. A *pdf-file* with the report and an *R Markdown file* containing the R-code.
2. A *pdf-file* with the report and a well commented *R-script*.

The pdf-file should be no longer than totally 15 pages (including everything). Include code in the text when you mean that it is informative for the reader; otherwise not. The R Markdown file or the R script should be possible to run if all R-packages used in the task is installed.

Be sure to clearly describe how you divide your data when you are using cross-validation to either determining tuning parameters or to evaluate predictions. Also, make sure that your variables are of the correct type, **numeric** for quantitative and **factor** for categorical variables.

The tasks

Start by installing the R-package **wooldridge**, Shea (2018) and load it. The dataset that you shall analyze is called **ceosal2** and contain information about the compensation of 177 CEO's. The dataset also contain 14 variables that might help in predicting CEO compensation. The R-help to the data contains the variable names. Your task is to predict the output variable **lsalary** (the logarithm of the CEO salary).

1. Describe relevant features of the output variable with descriptive statistics (tables and graphs). Also use descriptive statistics to investigate promising predictors for **lsalary**. Remove variables that cannot possibly be available when the salary of a new CEO should be predicted.
2. Use linear regression (on all or a subset of the predictors) and LASSO to predict **lsalary**. Evaluate the predictions on test data using an appropriate error measure.
3. Predict **lsalary** by KNN, using only **lsales** as a predictor. Determine K using leave-one-out cross-validation. Also, evaluate the predictions and compare with the results in 2.
4. Explain how you would use KNN if you had more than one predictor. Two alternative ways to do this have been mentioned in the course. If you would like to you can also write R-code to do this but this is not required to pass the assignment.
5. Argue for which, if any, variables might have a nonlinear relationship with **lsalary** and find a good model within the generalized additive models (GAM) framework. Evaluate the predictions.

References

Shea, Justin M. 2018. *Wooldridge: 111 Data Sets from "Introductory Econometrics: A Modern Approach, 6e" by Jeffrey m. Wooldridge*. <https://CRAN.R-project.org/package=wooldridge>.